# Introduction to Probability

Department of Computer Science

August 7, 2026



# Learning Objectives

- Experiment (Random Process)
- Calculating the Probabilities of : How it is Likely?
- Introduction to Conditional Probability

# Experiment (Random Process)

- Consider an experiment being performed
- The experiment will have one or more outcomes (**random process**).
  - A **sample space** is the set of all possible outcomes of a random process or experiment
  - An **event** is a subset of a sample space.

# Equally Likely Probability Formula

## Definition

If $S$ is a finite sample space in which all outcomes are equally likely and $E$ is an event in $S$, then the **probability of** $E$, denoted **P(E)**, is

$$P(E) = \frac{N(E)}{N(S)}$$

Where:

- N(E) is the number of outcomes in E
- N(S) is the total number of outcomes in S
- P(E) is between 0 and 1
- P(E)=0, the event will never occur
- P(E)=1, the event will always occur

# Calculating the Probabilities of : How it is Likely?

## Example

Suppose I flip two identical coins.

**What is the probability that I get two heads?**

**Solution:** There are four equally likely outcomes.

| Nickle | Dime |
|---|---|
| H | H |
| H | T |
| T | H |
| T | T |

$$P(HH) = \frac{N(HH)}{N(S)} = \frac{1}{4}$$

# Calculating the Probabilities of : How it is Likely?

## Example

Suppose I have a 50-50 chance of getting through a certain traffic light without having to stop. I go through this light on my way to work and again on my way home.

**Solution:** There are four equally likely outcomes.

| To Work | To Home |
|---|---|
| Stop | Stop |
| Stop | Don't Stop |
| Don't Stop | Stop |
| Don't Stop | Don't Stop |

- What is the probability of having to stop at this light at least once on a workday?
- What is the probability of not having to stop at all?

# Solution

What is the probability of having to stop at this light at least once on a workday?

$\Rightarrow$ Possible outcomes:(Stop, Stop) (Stop, Don't Stop) and (Don't stop, stop)

$\Rightarrow$ $P(S) = \frac{3}{4}$

What is the probability of not having to stop at all?

$\Rightarrow$ Possible outcome: (Don't Stop, Don't Stop)

$\Rightarrow$ $P(S) = \frac{1}{4}$

# Probability of an Event not occurring

## Probability of event not occurring=1 - Probablity of event occurring

## Example

There are several classes of CSI213 offered. There are some CSI213 Lecturers I like and some I don?t. I enroll in a class of CSI213 without knowing the Lecturer. A friend of mine has calculated that the probability that I get a Lecturer I like is: $P(L) = \frac{5}{9}$
**What is the probability that I will get a teacher that I don?t like?**

## Solution

$=>$ The $P(DL) = 1 - P(L) = 1 - \frac{5}{9}$

# Probability of an Event not occurring

## Probability of event not occurring=1 - Probablity of event occurring

- **Example:** Suppose we toss a pair of standard six-sided dice.

1. What is the probability that a we get a 7?
2. What is the probability that we get any sum but 7?

- **Solution:**

| Red die | Green die |
|---|---|
| 1 | 6 |
| 2 | 5 |
| 3 | 4 |
| 4 | 3 |
| 5 | 2 |
| 6 | 1 |

1. Probability of a 7

$$= \frac{6}{36} = \frac{1}{6} = 0.17 = 17\%$$

2. Probability of event **not** getting a 7

$$= 1 - \frac{1}{6} = \frac{5}{6} = 0.83 = 83\%$$

# Probability of General Union of Two Events

If $S$ is any sample space and $A$ and $B$ are any events in $S$, then

- The **disjunction** is the event that either A or B occurs. The probability of this disjunction:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

- **Example:** Suppose a librarian has a cart with 10 paperback algebra books, 15 paperback biology books, 21 hardbound algebra books, and 39 hardbound biology books. What is the probability that a book selected at random from this cart is an algebra book or a paperback book?

# Probability of General Union of Two Events

- **Solution:**

  Let A be an algebra book and B be a paperback book.

  Three probabilities: $P(A)$, $P(B)$, and $P(A \text{ and } B)$.

  Altogether, there are $10+15+21+39=85$ books.

$$P(A) = \frac{31}{85}, \quad P(B) = \frac{25}{85}, \quad P(A \text{ and } B) = \frac{10}{85}.$$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$
$$= \frac{31}{85} + \frac{25}{85} - \frac{10}{85} = \frac{46}{85} = 0.54 = 54\%$$

# Conditional Probability

A **conditional probability** is the probability that one event occurs given that another has occurred.

## Example

The accompanying table of data is adapted from a study of a test for TB among patients diagnosed with extra pulmonary TB.

| | HasTB | Does not have TB |
|---|---|---|
| Test Positive | 400 | 13 |
| Test Negative | 67 | 353 |

**Calculate the conditional probability that a person tests positive given that the person has TB.**

# Solution

Calculate the conditional probability that a person tests positive given that the person has TB

$\Rightarrow$ let $NUM_{TB}$ be all who have TB $= 400 + 67 = 467$

$\Rightarrow$ let $TP$ be true positives $= 400$

$\Rightarrow$ let $P(PT/TB)$ be probability that a person tests positive given that the person has TB.

$\Rightarrow$ $P(PT/TB) = \frac{TP}{NUM_{TB}} = \frac{400}{467}$

A generalization of this observation forms the basis for the following definition

# Definition: The probability of $B$ given $A$

Let $A$ and $B$ be events in a sample space $S$, if $P(A) \neq 0$, then the **conditional probability of B given A,** denoted $P(B|A)$, is:

$$\text{If } P(A) > 0,\ P(B|A) = \frac{P(A \cap B)}{P(A)}$$

# Example: The probability of $B$ given $A$

## Example

A pair of fair dice, one blue and the other gray, are rolled. What is the probability that the sum of the numbers showing face up is 8, given that both of the number are even?

## Solution

$\Rightarrow$ Using the product rule, the sample space is the set of all 36 outcomes from rolling the two dice.

$\Rightarrow$ Let $A$ be the event that both numbers are even and $B$ the event that the sum of the numbers is 8.

$\Rightarrow$ Then
$A = \{(2,2),(2,4),(2,6),(4,2),(4,4),(4,6),(6,2),(6,4),(6,6)\}$,
$B = \{(2,6),(3,5),(4,4),(5,3),(6,2)\}$ and
$A \cap B = \{(2,6),(4,4),(6,2)\}$

# The probability of $B$ given $A$

$\Rightarrow$ Because the dice are fair (so all outcomes are equally likely), $P(A) = \frac{9}{36}$, $P(B = \frac{5}{36})$ and $P(A \cap B) = \frac{3}{36}$

Example

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{\frac{3}{36}}{\frac{9}{36}} = \frac{3}{9} = \frac{1}{3}$$

# Example: The probability of $A$ given $B$

## Example

You roll two fair 6 sided dice. What is the probability that the first dice is 6 and the sum of two rolls is 9.

## Solution

$\Rightarrow$ Using the product rule, the sample space is the set of all 36 outcomes from rolling the two dice.

$\Rightarrow$ Let $A$ be the event the first dice is 6 and $B$ the event that the sum of two rolls 6.

$\Rightarrow$ Then $A = \{(6,1),(6,2),(6,3),(6,4),(6,5),(6,6)\}$,
$B = \{(3,6),(4,5),(5,4),(6,3)\}$ and $A \cap B = \{(6,3)\}$

$\Rightarrow$ Because the dice are fair (so all outcomes are equally likely),
$P(A) = \frac{6}{36}$, $P(B) = \frac{4}{36}$ and $P(A \cap B) = \frac{1}{36}$

# The probability of $A$ given $B$

**Example**

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{1}{36}}{\frac{4}{36}} = \frac{1}{4}$$

# Example: The probability of $A$ given $B$

## Example

You toss a fair coin three times. What is the probability that you get more Head (H) than Tail (T) given that the first toss is H.

## Solution

=> Using the product rule, the sample space is the set of all 8 outcomes from tossing a coin three times.

=> Let $A$ be the event that you get more H than T and $B$ the event that the first toss is H.

=> Then $A = \{(H,H,T),(H,T,H),(H,H,H),(T,H,H)\}$,,
$B = \{(H,H,H),(H,H,T),(H,T,H),(H,T,T)\}$ and
$A \cap B = \{(H,H,T),(H,T,H),(H,H,H)\}$

=> Because the toss are fair (so all outcomes are equally likely),
$P(A) = \frac{4}{8}$, $P(B) = \frac{4}{8}$ and $P(A \cap B) = \frac{3}{8}$

# The probability of $A$ given $B$

**Example**

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{3}{8}}{\frac{4}{8}} = \frac{3}{4}$$